

PopScript Documentation

A Comprehensive Guide to Syntax and Keywords

1. Introduction

PopScript is a statically-typed, interpreted programming language designed for simplicity, memory management experimentation, and rapid UI development.

2. Data Types

Keyword	Description	Example
<code>int</code>	64-bit integers	<code>int age = 25</code>
<code>float</code>	64-bit floating point numbers	<code>float price = 19.99</code>
<code>string</code>	Textual data	<code>string name = "Pop"</code>
<code>bool</code>	Boolean (true/false)	<code>bool ok = true</code>
<code>list</code>	Dynamic arrays of values	<code>list items = [1, "two", 3.0]</code>

3. Variables & Assignment

Variables must be declared with a type. Reassignment does not require the type keyword.

```
$/ Variable declaration
int count = 1
string label = "Items: "

$/ Reassignment
count = count + 1
```

4. Control Flow

Conditional (if)

Blocks are terminated with `stop;`.

```
if count > 0;
    print("Positive")
stop;
```

Loops (when & for)

`when` is used for conditional loops (while-style), and `for` is used for iteration.

```
when x < 10;
    x = x + 1
stop;

$/ Iterating over a range
for i with num(5);
    print(i)
stop;

$/ Iterating over a list
for item with myList;
    print(item)
stop;
```

5. Functions

Functions are defined with `func`. Use `export` to make them accessible to other scripts.

```
func greet(name);  
    print("Hello, ", name)  
stop;  
  
export func add(a, b);  
    return a + b  
stop;
```

6. Memory Management (ARMemory)

A unique feature of PopScript is Automatic Reference Memory (ARMemory). Enable it at the top of the file.

```
armemory = true  
  
alloc(myVar, 1024) $/ Allocate buffer  
free(myVar)       $/ Manual release
```

7. Modules and Libraries

Import built-in or downloaded libraries using `lib import`.

```
lib import ui  
lib import random  
  
int r = random.number(from=1, to=100)
```

8. Built-in UI Module

The `ui` module allows creating windows and widgets easily.

```
ui.window(title="My App", width=400, height=300)
ui.label(text="Main Screen")
ui.button(text="Click Me", action=myFunc)
ui.run()
```